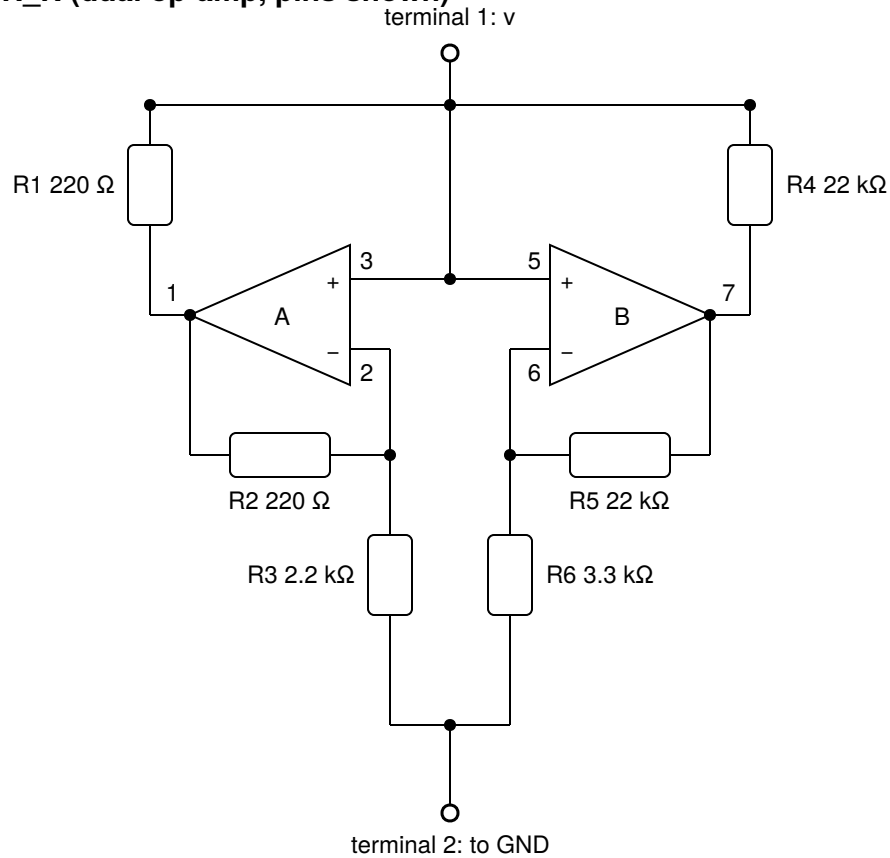
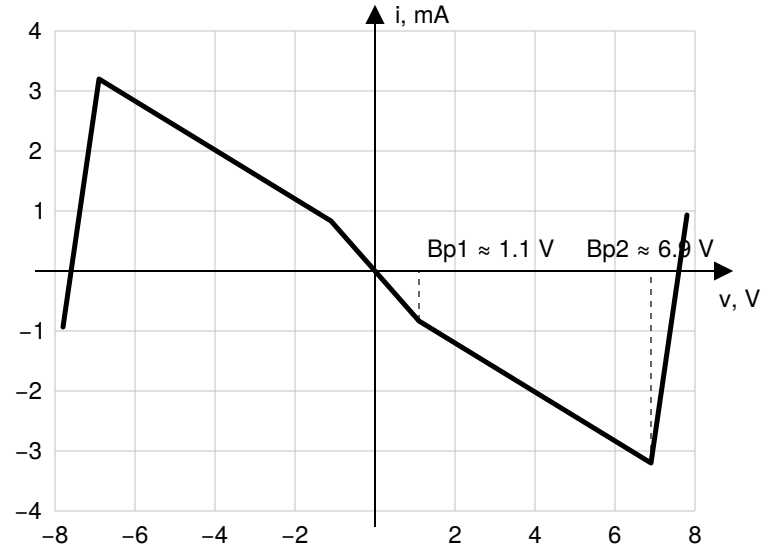


Chua diode N_R (dual op-amp, pins shown)



Expected V-I characteristic $i(v)$



$R_a = -(R_3 \parallel R_6) \approx -1.32$ k Ω (inner segment, $|v| < B_{p1}$)

$R_b = -R_3 \cdot R_4 / (R_4 - R_3) \approx -2.44$ k Ω (middle segments)

outer segments (op-amp saturation): $\approx +R_1 \parallel R_4 \approx +220$ Ω

Supply decoupling (TL082, shared pins 8 / 4)

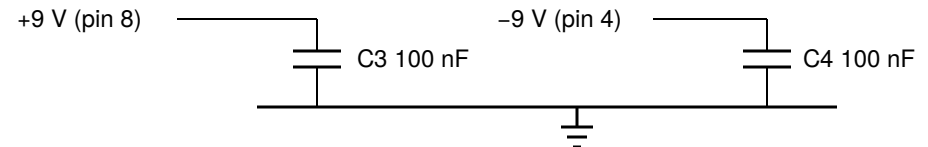


Fig. 1. Chua diode as a two-terminal element: circuit, expected piecewise-linear V-I curve, and supply decoupling.

Chua oscillator (N_R shown as a two-terminal element)

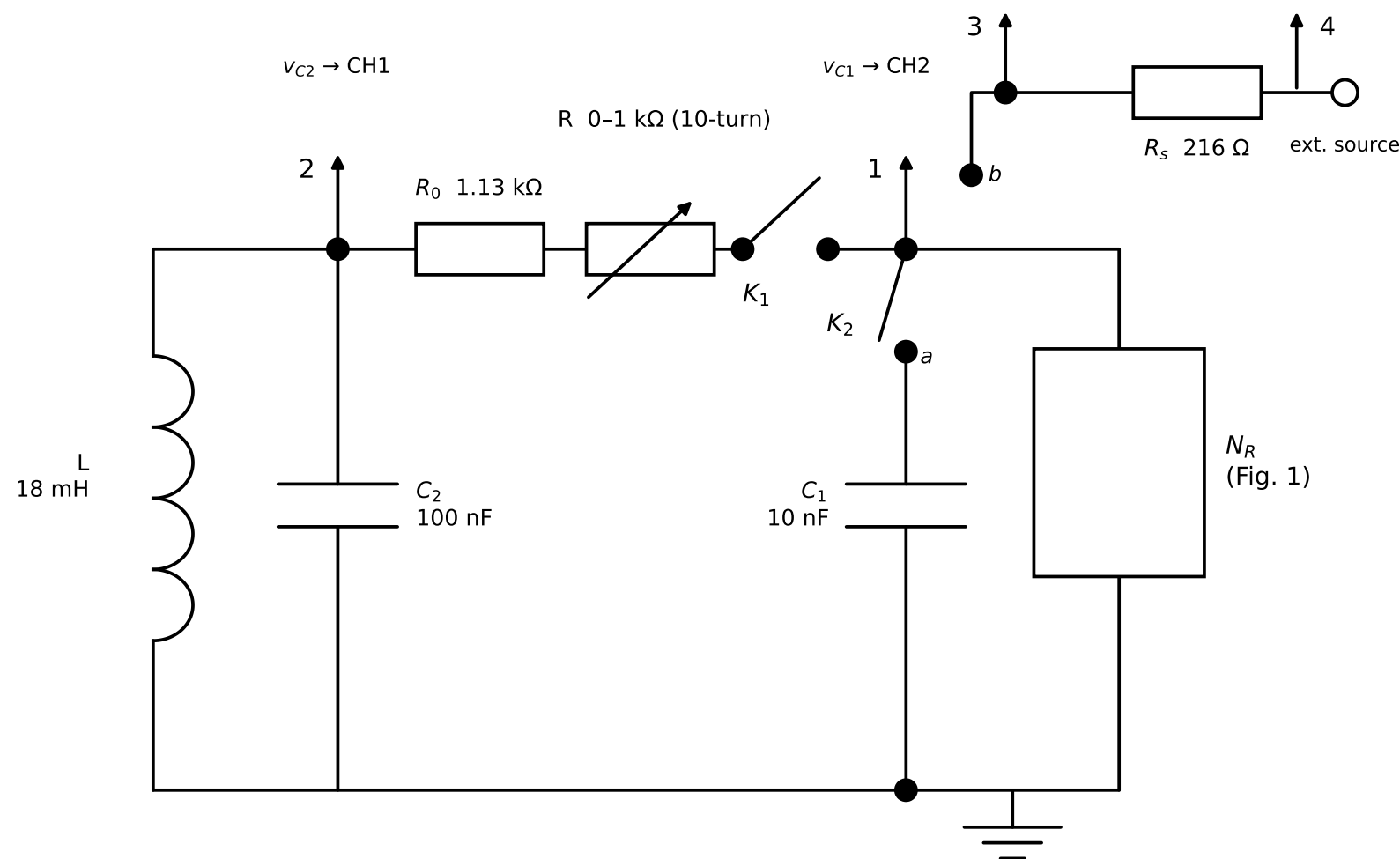


Fig. 2. Chua oscillator: resonant part (L , C_2), coupling resistance $R_0 + R$ with disconnect switch K_1 , C_1 , and the nonlinear element N_R of Fig. 1. K_2 in position a — oscillator mode; position b — V-I measurement of N_R (with K_1 open): a slow triangular sweep is applied at terminal 4, $v = V_3$ is the voltage across N_R and $i = (V_4 - V_3)/R_s$ its current. Probe points: 1 = v_{C1} , 2 = v_{C2} , 3 and 4 across the shunt R_s . Coupling resistance $R_{tot} = R_0 + R = 1.13...2.13$ kΩ; measurement points 1.54 / 1.74 / 1.91 kΩ. Dimensionless parameters: $\alpha = C_2/C_1 = 10$, $\beta = R_{tot}^2 C_2/L = 13...20$.